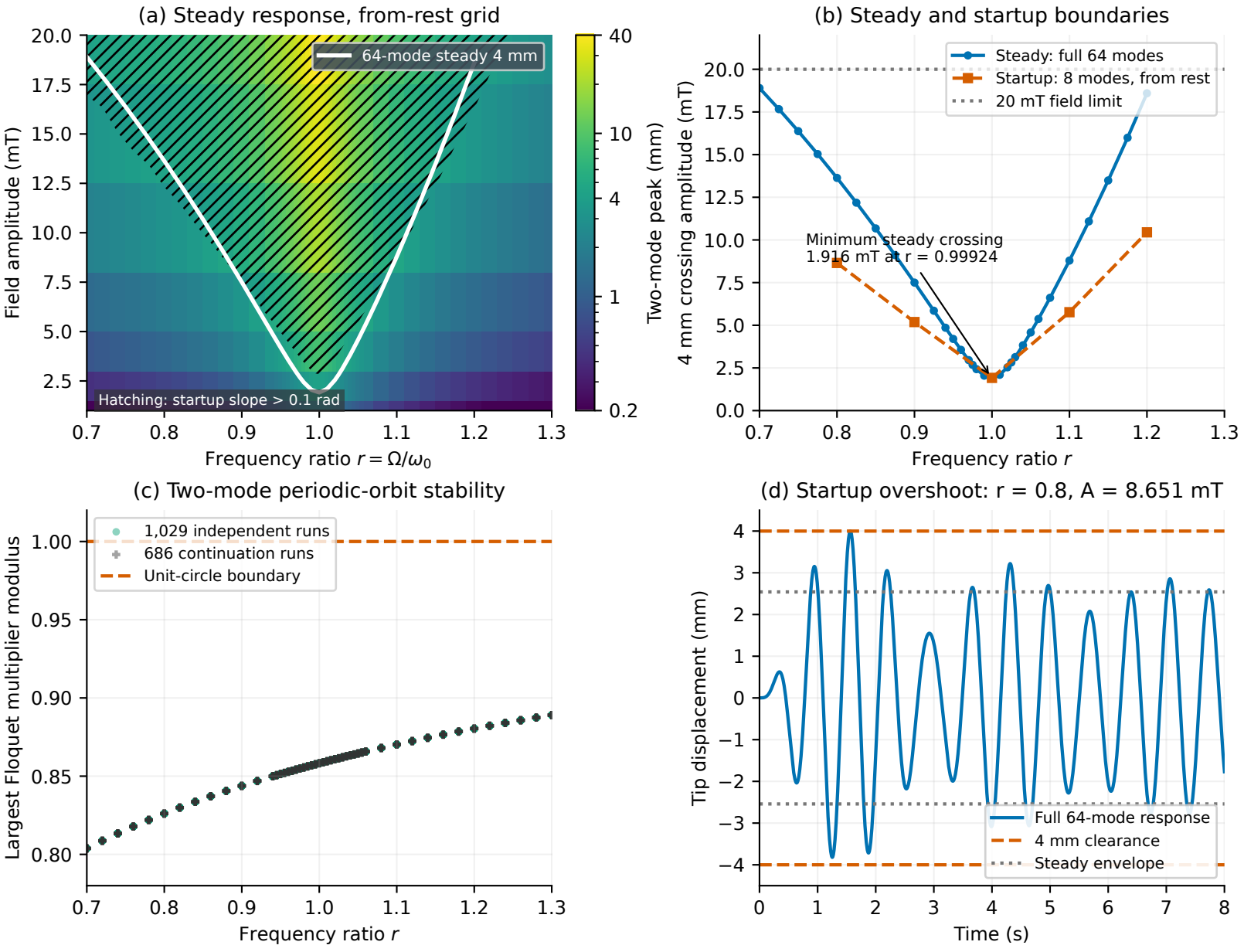
(a) Steady response, from-rest grid
64-mode steady 4 mm
Hatching: startup slope > 0.1 rad
Field amplitude (mT)
Frequency ratio $r = \Omega/\omega_0$
(b) Steady and startup boundaries
Steady: full 64 modes
Startup: 8 modes, from rest
20 mT field limit
Minimum steady crossing
1.916 mT at r = 0.99924
Two-mode peak (mm)
4 mm crossing amplitude (mT)
Frequency ratio r
(c) Two-mode periodic-orbit stability
1,029 independent runs
686 continuation runs
Unit-circle boundary
Largest Floquet multiplier modulus
Frequency ratio r
(d) Startup overshoot: r = 0.8, A = 8.651 mT
Full 64-mode response
4 mm clearance
Steady envelope
Tip displacement (mm)
Time (s)